Guidance on the Group Project

CA01 Investment Fundamentals, Autumn 2026

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The project is one sentence: **present a trading strategy and convince me of its merits.** That is the whole assignment, and it is deliberately open. This document is not a second set of instructions; it explains what "convince me" means, which tools you have, where the data are, and how the work is marked. Read it before you choose a topic.

All of it is guidance, not a specification. Strategies differ, and some of what follows will not fit yours. A measure that makes no sense for your strategy should not appear in your report, and I will not look for it. If your case is better made some other way, make it that way.

1. The task

You choose the strategy. It may be the case for owning a single company, a long–short portfolio, a factor tilt, an event trade, an options overlay, a prediction-market strategy, or anything else you can defend. There is no approved list and no answer I am looking for.

What is fixed is the standard of proof. You must show, with real data and the tools of this course, what the strategy earned and what risk it took, and you must compare it with what an investor could have done instead.

I mark the quality of the argument, not the sign of the return. A strategy you tested honestly and then rejected can earn a higher mark than one you talked up. Deciding *not* to invest is a perfectly good conclusion, provided you reached it with evidence.

2. What you hand in

- One report per group, **no more than 10 pages**, excluding references and appendix. Write it for an investor, not for me: someone deciding whether to put money behind you.
- An **appendix** of any length, plus the **code or spreadsheet that reproduces your numbers**. Neither counts toward the page limit. The appendix is consulted, not marked; the code exists so that any number in the report can be re-derived on request.
- Every table and figure labelled and referred to in the text. No raw software output pasted into the report.
- Format: 11pt type, single spacing, margins of at least 2 cm. Tables and figures count inside the ten pages; references and the appendix do not.
- The code must **run**. Submit it with the data, or with a script that fetches the data, so that someone else can reproduce your results without editing file paths.
- Groups: your study group, already established and fixed, with no swapping. Submit one report per group to Canvas. The deadline is **Monday, 19 October 2026 at 23:59**, and no late submissions will be accepted. If you hit a submission problem, email the report and the code to `rgomezcram@london.edu` *before* the deadline. If circumstances outside your control stop you submitting, use the School's extenuating-circumstances process rather than writing to me afterwards.
- Weight in the final grade: **30%**.

3. What "convince me" means

Every strategy report answers the same question:

> *Does this earn more than the alternative I would otherwise hold, for the risk it adds?*

Everything in your report exists to answer that question, and **no** is a perfectly good answer if that is where the evidence goes. What has to be convincing is the answer, not the verdict. An answer of either kind is credible when it has three things:

(i) **A mechanism.** An economic reason the return exists and has not been competed away. In practice this means naming who is on the other side of your trade and why they keep losing: a risk they are paid to avoid, a constraint they face, a mistake they repeat, or something you know that they do not. A story that would fit any result you might have found is not a mechanism: yours should imply something checkable in your own data, such as where the return should be strongest or when it should fail, and the report is much stronger for testing one of those implications.

(ii) **A magnitude.** A number, with units, over a stated period.

(iii) **Survivability.** Evidence that the number is not an artefact of the window you chose, the securities that happen to have survived, or the number of strategies you tried before this one worked.

The same finding can be reported well or badly. Here is one claim written five ways, from the weakest version to the strongest:

1. "This company is well managed." (an assertion)
2. "It returned 31% last year." (a number with nothing to compare it to)
3. "16.8% a year at 17.0% volatility, against 11.4% and 15.2% for the market, over 2016–2026." (a comparison)
4. "…and an alpha of 5.4% a year, $t = 2.2$." (risk-adjusted)
5. "…and it held up in the half of the sample we did not use to design it, and here is why the return exists." (out of sample, with a mechanism)

A report that stops at rung 2 will not pass. A report that reaches rung 5 will do well even if its conclusion is "do not invest".

4. The tools, and where they come from

You decide which measures make your case. Nothing below is compulsory, and no report needs all of it. But a claim about performance that arrives without risk beside it is not an argument, and I will read it as one.

Drawdown, turnover and trading costs arrive in Lecture 8 and are defined below, so you can plan around them. Options and prediction markets arrive in Lectures 9 and 10, the last week before the deadline, and are not defined here: choose one of those families only if you already know the instrument, and come and see me early if you do.

Tool	Comes from
Discounting, present value, compounding	Lecture 1
Bond prices, yields and forward rates	Lecture 2
The yield curve, duration, interest-rate risk	Lecture 3
Equity valuation: dividend discount, Gordon, multiples	Lecture 4
Returns: cumulative, annualised, arithmetic vs geometric	Lecture 5
Volatility, correlation, diversification	Lecture 5
Sharpe ratio; what an asset adds to a portfolio	Lecture 6
Beta, CAPM alpha	Lecture 7
Multifactor alpha, factor loadings, style	Lecture 8
Maximum drawdown, turnover, trading costs	Lecture 8
Option payoffs, replication, put–call parity	Lectures 9–10
Prediction markets	Lectures 9–10

For each tool: the formula, the question it answers, and the mistake I see most often.

Tool	Formula	The mistake
Returns *What did £1 become, and at what rate did it grow?*	cumulative $\prod_t(1 + r_t) - 1$; annualised $[\prod_t(1 + r_t)]^{f/T} - 1$, with $f = 12$ monthly, 252 daily	Reporting the arithmetic mean as what the strategy delivered. As Lecture 5 puts it: the geometric mean measures past performance, the arithmetic mean forecasts future performance. Never compare cumulative returns over windows of different lengths.
Volatility *How wide is the range of outcomes?*	$\sigma = \sqrt{\frac{1}{T-1}\sum_t (r_t - \bar{r})^2}$; $\sigma_{\text{ann}} = \sqrt{12}\,\sigma_{\text{month}} = \sqrt{252}\,\sigma_{\text{day}}$	Annualising the mean by $\times 12$ and the volatility by $\times 12$ as well. Mixing daily and monthly statistics in one table.
Sharpe ratio *How much excess return per unit of total risk?*	$S = (\mathrm{E}[r_P] - r_f)/\sigma_P$	Leaving out r_f, which is four to five points at today's rates, or subtracting it twice from a self-financing long–short book, whose return is already an excess return. Borrowing at r_f to scale a position up leaves S unchanged: a bigger return at proportionally bigger volatility is leverage, not skill.
Beta and alpha *Did it beat what its market exposure alone would have earned?*	$r_P - r_f = \alpha + \beta\,(r_m - r_f) + \varepsilon$	Regressing raw rather than excess returns; quoting a monthly α as annual; reporting α with no t-statistic and no sample period. Rule of thumb: $t \approx \mathrm{IR}\sqrt{\text{years}}$, where $\mathrm{IR} = \alpha/\sigma_\varepsilon$ annualised. So IR = 0.5 needs about sixteen years to reach $t = 2$. Short samples rarely prove anything.
Multifactor alpha *What is left after the risks a reader can buy cheaply?*	$r_P - r_f = \alpha + \beta_{\text{MKT}}\text{MKT} + \beta_{\text{SMB}}\text{SMB} + \beta_{\text{HML}}\text{HML} + \beta_{\text{MOM}}\text{MOM} + \varepsilon$	Running several models and reporting only the one with the largest α; hiding the loadings, which are often the most interesting result. "Our stock-picking is a 0.8 loading on momentum" is a finding.
Maximum drawdown *Worst loss sat through, and how long to make it back?*	$V_t = \prod_{s \le t}(1 + r_s)$, $M_t = \max_{s \le t} V_s$, $D_t = V_t/M_t - 1$; $\mathrm{MDD} = \min_t D_t$	Computing it on returns instead of on the value index; quoting it without dates, without recovery time, or without the same number for the benchmark.

Tool	Formula	The mistake
Turnover *How much does the strategy trade?*	$\tau_t = \frac{1}{2}\sum_i \lvert w_{i,t} - w_{i,t^-}\rvert$; net of a one-way cost c, $r^{\text{net}} = r^{\text{gross}} - c\,(2\tau)$	Presenting a strategy that trades constantly as though trading were free. See the note below on how far I expect you to take this.
Correlation with what the reader owns *Does this improve a portfolio I already hold?*	$\rho_{PB} = \sigma_{PB}/(\sigma_P\sigma_B)$	Judging a diversifier by its own Sharpe ratio: adding a *small* allocation of P raises B's Sharpe ratio if and only if $S_P > \rho_{PB}S_B$, so a low correlation can earn a place for a mediocre asset. Quoting a full-sample correlation without checking what it was in the months B lost the most.

Lecture 6 assumes no trading costs so that the geometry stays clean; in practice they are there. **I am not expecting you to compute returns net of transaction costs.** Costs are genuinely hard to measure, and a confident number is usually a made-up one. What I do expect is a **comment**: say roughly how much your strategy trades, and whether plausible costs would leave the result standing. If you want to go further, the neatest way is the *break-even cost*, the level of trading costs at which the strategy stops working. A paragraph of honest reasoning is enough. Prediction markets are the exception: there the fees are published, so use them.

5. Choosing what to compare against

A performance number on its own is not an argument. The benchmark is the best thing the reader could have done instead, with the same money, at the same risk , not "the market" by default. Report the same statistics for it, over identical dates.

Three comparisons are worth making, and most reports need at least two:

- **The naive alternative:** buy and hold the index, or hold your own universe in equal weights. Equal weighting is much harder to beat than students expect.
- **The risk-matched alternative:** the index scaled to your strategy's volatility. Scale the *excess* return, $r_f + k\,(r_m - r_f)$ with $k = \sigma_P/\sigma_m$, not the total return. Where your volatility differs materially from the benchmark's, this is the comparison you cannot skip.
- **The style-matched alternative:** if your strategy is a value strategy, compare it with value; if it is small-cap, compare it with small caps.

The choice of benchmark is itself a claim, so defend it in a sentence. If your strategy is long UK small caps and you benchmark it against cash, you have not shown skill. You have shown that equities beat cash.

6. In sample and out of sample

The distinction is not about dates. **Out of sample means data that had no influence on any choice you made**: the universe, the window, a parameter, a threshold. If you looked at it before fixing the rule, it is in sample, whenever it happened.

Lecture 6 makes the point with the course's own data. Estimate tangency weights on ten stocks over five years and the optimised portfolio posts a Sharpe ratio of **1.93** in the sample it was fitted on. Hold those same weights for the next five years and it collapses to **0.24**, while equal weighting

the same ten stocks holds at **1.35**. Nothing about the stocks changed; only whether the data had been used to choose the weights.

Three honest ways to handle this in a short project:

- Split your sample, design on the first half, and report both halves.
- If the sample is too short to split, say so, and report performance in the worst sub-period rather than only the full-sample average.
- Report the whole search, not the survivor: a result that exists at only one parameter value is not a result.
- If your verdict is that the strategy does not work, make sure the test could have found the effect had it been there. Report the standard error beside the estimate, and say what size of return your sample could have distinguished from zero. "We found nothing" from three years of monthly data is not a finding.

7. A menu of strategies

This is a menu, not a list of approved topics. Each entry names what makes that family hard to argue well.

- **A single company, or a few of them: a concentrated portfolio.** Value them with Lecture 4, and any other lecture that fits, and state what today's price must already be assuming. Convincing when your recommendation follows from a disagreement about one identifiable input (growth, payout, the discount rate) and you can say *why* the consensus is wrong on it. Lecture 4 supplies that evidence: long-horizon growth forecasts are systematically too optimistic and extrapolate the recent past, while the short-horizon number is managed, so almost every firm beats it. The bias does not run the same way in every sector, so check rather than assume. A case built on a documented bias beats one built on your own guess about growth. If you hold several names, say what the concentration costs you: Lecture 5 shows how much idiosyncratic risk a handful of stocks still carries. *Trap:* describing the companies well and never confronting their prices.
- **Long–short.** Rank a universe on a signal, buy the top, sell the bottom. Convincing when the ranking rule is mechanical and fixed before you look at results, the two legs are reported separately, and borrow cost, gross exposure and turnover are all stated. *Trap:* not saying what capital your return is a return *on*.
- **Pairs.** Two economically linked securities whose price ratio has been stable. Convincing when the economic link comes first and the statistics second, and when you report the times the spread did not converge. *Trap:* mining many pairs and showing the one that worked.
- **Bonds and the curve.** A view on the level of rates, a steepener or flattener on the slope, or a carry position that earns the roll-down. Convincing when you state the position in **duration** terms, price it off the curve rather than off a single yield, and show separately what a parallel shift and a twist do to it, using the tools of Lectures 2 and 3. *Trap:* reporting a yield as though it were a return. The return on a bond position is carry plus price change, and duration is what turns a rate move into the second part.
- **A factor tilt.** Overweight value, size, quality or low volatility. Convincing when measured against the factors a reader can already buy for a few basis points. *Trap:* rediscovering HML and calling it alpha.

- **Sector rotation.** Move weight between sectors on a rule. Convincing when the signal was observable on the day you would have traded. *Trap:* using macroeconomic data as revised today rather than as first published.
- **Momentum or reversal.** Convincing when formation and holding windows are stated, turnover is measured, and the crash risk is shown. *Trap:* ignoring costs. Momentum trades fast, and costs, not returns, usually decide the answer.
- **Event-driven.** Earnings, buybacks, index additions, takeovers. Convincing with a clean event list, a stated window, and mean *and* median returns reported with the number of events. *Trap:* too few events to detect anything. Count first.
- **An options overlay.** A covered call, a protective put, a collar. Convincing when you show the payoff, price the position, and report the effect on the *whole portfolio's* distribution rather than as a standalone return. *Trap:* presenting premium income as return. A covered call sells the upside, and what it gives up is the entire question.
- **Prediction markets.** Binary event contracts on Kalshi or Polymarket. Convincing when you follow a rule rather than a view: enforcing the law of one price across mutually exclusive outcomes, taking the other side of a known bias, or reacting faster to scheduled news, at prices actually quoted. These contracts settle at 0 or 1, so a monthly return series and a market regression are the wrong frame: report the **profit and loss** and the **return on the capital you put up**, bet by bet and in total, and use the exchange's own published fees. If your contracts carry a systematic exposure to rates, an election or an inflation print, say so rather than assume they do not. *Trap:* zero-sum before fees and negative-sum after, so you must say who is on the other side and why they lose.
- **A diversification overlay.** Add gold, a commodity or bitcoin to a portfolio the reader already holds. The test is not the new asset's own Sharpe ratio but whether it raises the combination's. *Trap:* a full-sample correlation with no look at the months that mattered.

If your recommendation is one of the ten largest companies in the world, you are arguing against a very well-informed consensus. Say what you know that it does not.

8. Data

You source your own data. Being resourceful about it is part of the exercise.

- **WRDS** is the serious option: CRSP, Compustat and much else. You should have access through the School. Email the library to have your account set up, and do it early rather than the week before the deadline.
- **Kenneth French's data library** has the market, SMB, HML, momentum, industry portfolios and the risk-free rate, free, in monthly and daily form. This is where the course's own factor data come from. Take the factors from the same region as your strategy. The library has separate US, European and global files, and a UK strategy benchmarked on US factors is not benchmarked.
- **Yahoo Finance** is fine for prices and quick work. Compute returns from the adjusted close, which puts dividends back in; delisted companies are missing altogether, which is survivorship bias, so say so.
- **FRED** for rates and macroeconomic series; **SEC EDGAR** for filings, including 13F holdings; **Polymarket** and **Kalshi** publish price histories through public APIs.

- **FactSet** is available through an LBS subscription, and the **LBS library** has terminals, further subscription databases, and staff whose job is to help you find data. Book ahead.
- If you cannot find what you need, **come and see me** and I will point you in the right direction.

Four things you must state, and which no lecture will tell you:

- Whether your returns are **total or price** returns, and in **what currency**. A sterling investor's return on a US strategy includes the exchange rate.
- **Which risk-free rate** you used. Use one series throughout and name it.
- **Which frequency**, and stick to it. Monthly for anything held longer than a month; daily only if the strategy trades daily.
- Whether the data are **as they were then or as revised since**.

Before you commit to a topic, check you can actually get the data, say 60 monthly or 500 daily observations, in an afternoon. If a source requires a subscription you do not have, change the topic, not the standard of evidence. Do not spend more than a fifth of your time collecting data: a simple strategy analysed well beats an ambitious one you could not finish.

9. Six ways to lose the argument

1. **Look-ahead bias.** Rebalancing on 1 January using earnings that were not published until March. *Fix:* for every input, write down the date it became public, and lag it.
2. **Survivorship bias.** Testing "the ten largest UK banks" on today's list. Northern Rock and HBOS are not on it. *Fix:* use the universe as it stood, or state plainly that you have not and bound the effect.
3. **A cherry-picked window.** Any long equity strategy measured from March 2009 looks good. *Fix:* justify the start date and show performance year by year, so the reader can see which years carry the result.
4. **Overfitting.** "We tried 10-, 20- and 60-day windows and report the 20-day." *Fix:* report all three. The spread across variants is itself evidence.
5. **Leverage mistaken for skill.** "We returned 28% against the market's 12%". At 2.4 times the volatility, that is the market. *Fix:* volatility and Sharpe ratio beside every return.
6. **A t-statistic with no economic story.** *Fix:* name the counterparty and why they keep losing. If you cannot, say so. The admission is worth more than an invented story.

10. How to structure 10 pages

Section	Pages	Must contain
The recommendation, up front	0.5	the claim, the headline number, the verdict
The strategy, precisely stated	1.5	rules, universe, sizing, rebalancing
Why it should work	1.5	the mechanism, and who is on the other side
Data and sample	1	sources, dates, currency, r_f, frequency
Results	3	one results table, one or two figures
Robustness	1.5	out of sample, variants, the worst period
What would change my mind	1	the strongest case against

The table above is a rough allocation, not a required table of contents. Use the structure that suits your argument.

Put your strategy and its benchmarks side by side in a single table, on identical dates, with one row per statistic that bears on your claim. Return, volatility and the Sharpe ratio will almost always be among them; alpha, drawdown and turnover only where the argument turns on them. Do not add a net-of-cost row unless you can defend the cost assumption behind it. Section 4 asks for a paragraph, not a number. Most reports are carried by one table and two figures done properly. If a figure does not change what the reader believes, cut it.

11. How it is marked

Criterion	Weight
Specification of the strategy: could a reader run it?	15
Economic reasoning: why does the return exist?	20
Measurement: the tools applied correctly	25
Benchmarking and robustness	20
Honesty about weaknesses	10
Communication: within length, clear, nothing unreferenced	10
Total	**100**

The criteria are applied with judgement. Where one of them does not fit your strategy, for instance if there is no sensible benchmark or no alpha to estimate, you lose nothing by saying so and explaining why; you lose marks only for leaving a gap unaddressed.

- **A strong report** states a strategy precisely enough to run, gives a mechanism the reader believes, measures it correctly against a benchmark it defends, tests it outside the data used to build it and reports what it found either way, says plainly what trading it would cost, and makes the case against its own conclusion.
- **A good report** measures correctly against a fair benchmark and reaches a defensible verdict, but asserts the mechanism rather than arguing it, or leaves the robustness work half-done.
- **A passing report** states a strategy and applies some tools correctly, but the comparison is weak, the risk measures are thin, and the weaknesses go unmentioned.
- **A failing report** claims performance with no benchmark, no risk measure, or no data. An opinion with numbers attached.

None of these bands depends on whether the strategy made money. A well-executed report concluding that the strategy does not work belongs in the top band; a lucky back-test presented without a benchmark belongs in the bottom one.

12. Before you submit

- Could a reader reproduce your positions from your description alone?
- Is every performance number matched by the same number for a benchmark, over identical dates?

- Have you said something about what it costs to trade the strategy?
- Are Sharpe ratios computed on excess returns and annualised by $\sqrt{\text{time}}$?
- Does every alpha carry a t-statistic and a sample period?
- Have you said how many variants you tried?
- Does the drawdown come with dates and a recovery time?
- Is there a sentence naming who is on the other side of the trade?
- Does the report contain the strongest argument against your own recommendation?
- Is it within 10 pages, with every table and figure referred to in the text?
- **Can you re-derive every number in the report on request?** If not, remove it.